

Southeastern Section - 68th Annual Meeting - 2019

Paper No. 16-5

Presentation Time: 1:00 PM-5:00 PM

ULTRA HIGH-DEFINITION DIGITAL SURFACE MODEL AND GEOCHEMICAL MAPPING OF THE STEFANOS CRATER AT THE ACTIVE VOLCANO OF NISYROS IN GREECE WITH UAV SCANNING, PORTABLE X-RAY FLUORESCENCE AND GEOGRAPHICAL INFORMATION SYSTEMS

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The Hellenic Active Volcanic Arc (HAVA) has resulted from the interaction of the rolling-back and subducting African plate and the overriding and extending Aegean microplate. At the western end of the HAVA, Nisyros volcanic island hosts the Stefanos crater with a history of numerous volcanic eruptions and associated faulting. Drone mapping at an altitude of 30 meters above the crater's surface has generated a digital surface model (DSM) and cloud densification of approximately 40 million points with an average of 9601.62 per m² 3D densified points. An ultra high-resolution (UHD) of 1.05 cm per pixel DSM for a 34,000 m² coverage was developed. Post-processing of the images was conducted in PIX4D software, the acquirement of the images performed by an autonomous drone, DJI Matrice M100, with a Zenmuse X3 camera.

Aerial mapping was accompanied by Portable X-Ray Fluorescence (PXRF) analysis with a Niton XL3t GOLDD+ which was performed on a grid with 10 x 10 meters cells on the crater's surface, and 155 measurements were conducted for elemental analysis. Probing the Stefanos crater's surface with the PXRF has revealed an average content in parts per million (ppm) of (Mo) 4.5 ± 2.2, (Zr) 135.9 ± 3.5, (Sr) 213.5 ± 3.8, (U) 5.1 ± 3.5, (Rb) 27.8 ± 2.1, (Th) 9.0 ± 2.9, (Pb) 136.1 ± 7.1, (Se) 3.8 ± 2.1, (As) 37.5 ± 6.1, (Zn) 12.0 ± 5.2, (Cu) 23.0 ± 11.1, (Ni) 23.8 ± 11.5, (Co) 39.7 ± 16.9, (Fe) 6651.3 ± 91.2, (Mn) 68.3 ± 30.8, (Cr) 31.6 ± 6.8, (V) 42.0 ± 15.1, (Ti) 1679.6 ± 58.9, (Sc) 35.3 ± 12.6, (Ca) 3696.1 ± 98.2, (K) 5410.7 ± 149.4, (S) 8919.6 ± 351.3, (Ba) 658.5 ± 37.2, (Cs) 31.5 ± 7.5, (Te) 54.1 ± 23.3, (Sb) 24.5 ± 11.6, (Sn) 18.6 ± 10.7, (Cd) 17.6 ± 10.2, (Ag) 14.1 ± 7.4. Geographical information systems (GIS) analysis and implementation of inverse distance weight prediction mapping using the geostatistical analyst module has shown linear features of contrasting geochemical compositions implying the existence of faulting, possibly active and rooted in the upper hydrothermal system underneath the Stefanos crater.

Session No. 16--Booth# 5

[T12. Natural Hazards Prediction \(Posters\)](#)

Thursday, 28 March 2019: 1:00 PM-5:00 PM

[Carolina Ballroom \(Francis Marion Hotel\)](#)

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